

2x2 Between-Subjects Designs: Main Effects and Interactions

Instructions: In your group, complete the following questions about 2x2 Between-Subjects Designs.

1. “Imagine you are an educational psychologist who is examining how much anger students

after		Private	Public	experience
negative	Positive Feedback	2	2	receiving
After	Negative Feedback	5	7	feedback.
pseudo-				taking a

IQ test, you give participants one of two types of feedback (positive, negative) in one of two methods (privately, publicly). Afterwards, you measure their anger on a 7-point scale (1 = not very angry, 7 = very angry).” Below are the data.

Using the data above, answer the following questions:

- a) Is there a main effect for type of feedback?

There is a main effect for type of feedback, such that participants reported higher levels of anger in the negative feedback condition than the positive feedback condition.

- b) Is there a main effect for setting of feedback?

There is a main effect for setting of feedback, such the participants reported participants reported higher levels of anger in the public setting condition than in the private setting condition

- c) Is there an interaction between type of feedback and setting of feedback? Explain in APA format.

There is an interaction between type of feedback and setting. Although negative feedback increased anger in both private and public settings, the increase was larger when feedback was delivered publicly than privately.

2. “Imagine you are a cognitive psychologist who is examining the effects of music on memory recall. You

		1 minute	5 minutes	have
listen to	Classical Music	7	15	participants
classical	No music	7	15	either
music				music or no

while memorizing 20 faces for either 1 minute or 5 minutes. Afterwards, participants are tested to see how many faces they can accurately recall.” Below are the data.

Using the data above, answer the following questions:

- a) Is there a main effect for type of music? Explain in APA format.

There is not a main effect for type of music. Participants recalled the same number of faces in the classical music condition as the no music condition.

- b) Is there a main effect for duration of memorization? Explain in APA format.

There is a main effect for duration of memorization, such that participants in the 5 minute condition recalled more face than those in the 1 minute condition.

- c) Is there an interaction between type of music and duration of memorization? Explain in APA format.

There is no interaction between type of music and duration of memorization. The increase in faces recalled from the 1 minute condition to the 5 minute condition was the same across both types of music.

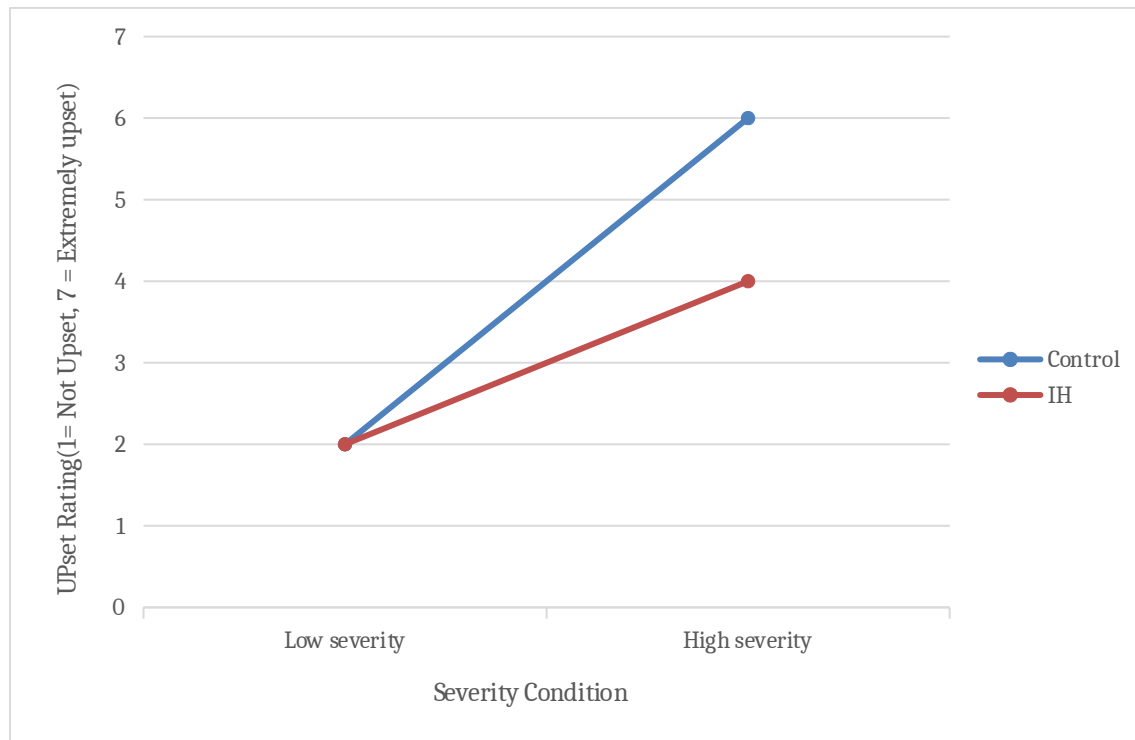
3. *“Imagine you are a psychologist who wants to examine the effects of intellectual humility (IH) and offense severity on forgiveness (i.e., from the victim’s perspective). Researchers randomly assigned participants to an intellectual humility condition (i.e., where they reflected on a time they acted with IH) or a control condition. Participants then recalled either a high severity or low severity offense committed against them. Researchers then measured how forgiving they were of the person who committed an offense against them (1=Not at all upset with the other person, 7=Extremely upset with the other person).*

Hypotheses:

- a) There will be a main effect for intellectual humility, such that those in the intellectual humility condition will report higher forgiveness than those in the control condition.
b) There will be a main effect for severity, such that participants in high severity condition will report lower levels of forgiveness than participants in the low severity condition.
c) There will be an interaction between IH and severity. Within the high severity condition, participants in the IH condition will report higher forgiveness than participants in the control condition. Within the low severity condition, we do not expect differences in forgiveness between the IH and control conditions.

Based on these hypotheses, write out the values in the table below to confirm these hypotheses. Additionally, draw out a graph to show a visual representation of the data (**Note:** You can input any values you choose, as long as they confirm the hypotheses).

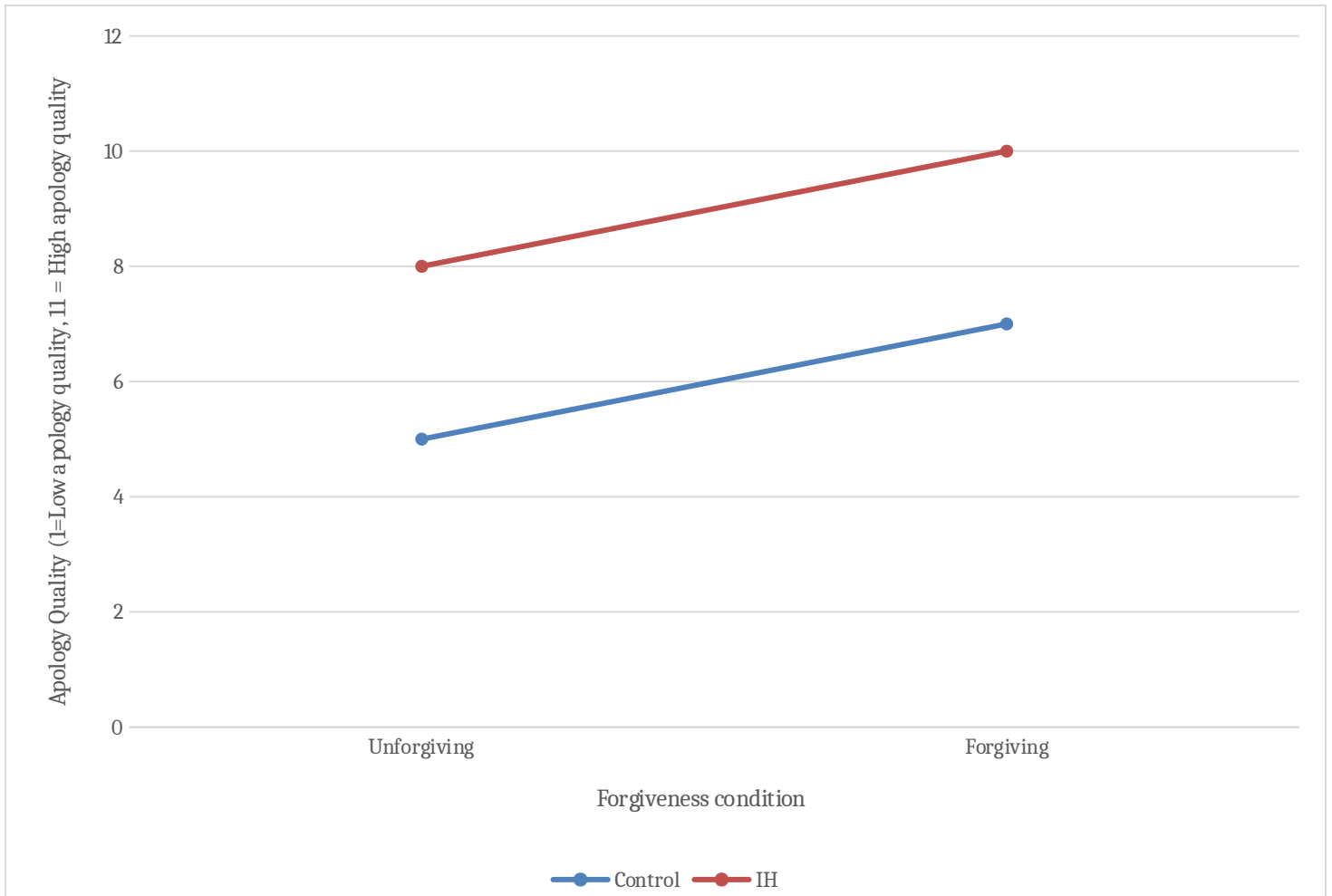
	Low severity	High severity
Control	2 (not very upset – very forgiving)	6 (very upset – not very forgiving)
IH	2 (not very upset – very forgiving)	4 (somewhat upset – somewhat forgiving)



Note that our experimental design follows the same 2 x 2 format. Below, write in the means that you might expect in each

	Unforgiving	Forgiving
Control	5	7
IH	8	10

condition. Draw a graph to illustrate the hypothesized results. *Note:* remember that we will measure relationship satisfaction on a 1 – 7 scale (1 = *Low apology quality*, 11 = *High apology quality*)



4. Finally, brainstorm why do you think this interaction will occur? In other words, why do you think the effect of intellectual humility might depend on the victim's post-transgression response? This will become clearer as you read the literature and outline and write your Introduction sections; however, for now, what is one reason to expect an interaction?

The effect of IH may depend on the victim's post-transgression response because of the victim's willingness to repair the relationship. While I already predicted that IH will have a pretty high quality apology in the unforgiving condition, I think that the more a victim is willing to repair (i.e. they are more forgiving), the higher the apology quality will be.